

CG Shift Calculation

Preliminary ARG27 longitudinal CG shift from the accumulator and inverter changes

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PRELIMINARY The ARG27 accumulator and inverter CAD were not built or placeable in the car at the time of this calculation, so their ARG27 positions are estimates. See the assumptions.

1 Context

For ARG27, the accumulator and inverter are being moved and both are getting lighter. We design the car to a 50/50 weight distribution, so if the longitudinal (front to rear) CG moves, we may have to shift the wheelbase to bring it back. This page calculates a preliminary estimate of that longitudinal CG shift.

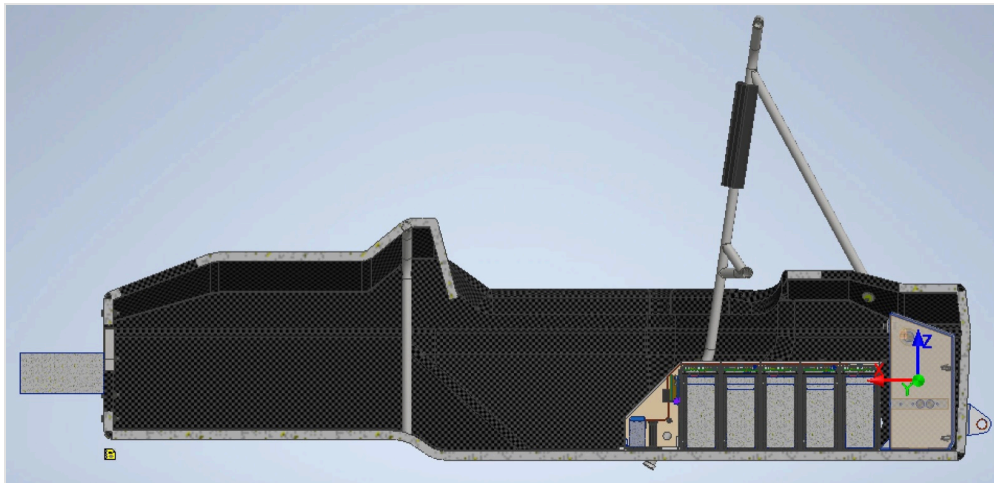


FIGURE 1 ARG26 packaging, side view.

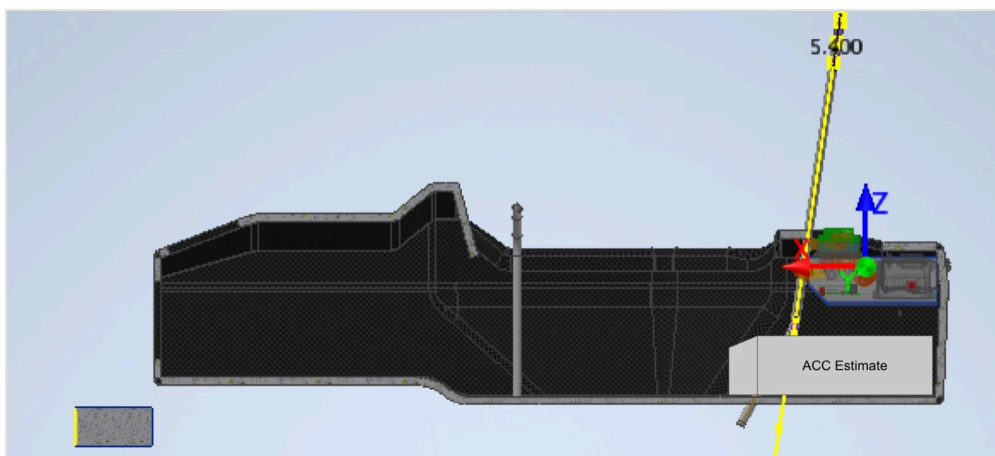


FIGURE 2 Preliminary ARG27 packaging, side view, with the accumulator shown as an estimate.

2 Methodology

Positions are x in vehicle coordinates, in inches, with $+x$ pointing forward. All masses are sprung masses, so the ARG26 total (m_{26}) is the total sprung mass of the car with the driver and x_{26} is its sprung CG.

2.1 Assumptions

- This calculation is preliminary. The ARG27 accumulator and inverter CAD were not built or placeable in the car at the time, so the ARG27 positions come from hand estimates instead of CAD.
- The accumulator CG shift can be modeled with a rectangular analysis: the accumulator is treated as a rectangle with its CG in the middle, so making the case longer moves the CG by half the added length.
- The inverter enclosure is the same box on both cars with its CG in the middle, so turning it from standing up to lying flat moves the CG by half the change in its length along x .
- The ARG27 accumulator and inverter masses are estimates from the HV lead.
- Everything other than the accumulator and inverter is lumped together as "other" and is the same on both cars.

2.2 Remove the ARG26 Accumulator and Inverter

$$m_{\text{other}} = m_{26} - m_{\text{ACC26}} - m_{\text{INV26}}$$

$$x_{\text{other}} = (m_{26} x_{26} - m_{\text{ACC26}} x_{\text{ACC26}} - m_{\text{INV26}} x_{\text{INV26}}) / m_{\text{other}}$$

2.3 Add the ARG27 Accumulator and Inverter

$$m_{27} = m_{\text{other}} + m_{\text{ACC27}} + m_{\text{INV27}}$$

$$x_{27} = (m_{\text{other}} x_{\text{other}} + m_{\text{ACC27}} x_{\text{ACC27}} + m_{\text{INV27}} x_{\text{INV27}}) / m_{27}$$

2.4 Longitudinal CG Shift

$$\Delta x_{\text{CG}} = x_{27} - x_{26}$$

A positive shift means the CG moved forward.

2.5 Effect on Weight Distribution

With wheelbase L , the shift moves this much of the weight onto the front axle:

$$\Delta(\text{front \%}) = 100 \times \Delta x_{\text{CG}} / L$$

3 Results

3.1 Inputs

Part	Mass (lb)	x (in)
ARG26 sprung mass with driver	470	-50.79
ACC26 (accumulator)	102	-64.00
INV26 (inverter)	34.17	-78.20
ACC27 (accumulator)	95	-69.04
INV27 (inverter)	16.53	-74.77

The ARG27 accumulator moves 5.04 in toward the rear, from $x = -64.00$ in to $x = -69.04$ in. The whole accumulator moves back 7.60 in to fill the space the inverter used to take up, and the case getting longer at the front moves its CG forward 2.56 in. The ARG27 inverter moves 3.43 in forward, from $x = -78.20$ in to $x = -74.77$ in. See the appendix for where each input comes from.

3.2 Change in Each Part

Part	Mass removed (lb)	x shift (in), + is forward
Accumulator	7.00	-5.04
Inverter	17.64	+3.43
Total	24.64	

3.3 Calculation

Step 1, everything else on the car:

$$m_{\text{other}} = 470 - 102 - 34.17 = 333.83 \text{ lb}$$

$$x_{\text{other}} = (470(-50.789) - 102(-64) - 34.17(-78.2)) / 333.83 = -43.947 \text{ in}$$

Step 2, ARG27:

$$m_{27} = 333.83 + 95 + 16.53 = 445.36 \text{ lb}$$

$$x_{27} = (333.83(-43.947) + 95(-69.04) + 16.53(-74.77)) / 445.36 = -50.444 \text{ in}$$

Step 3, CG shift:

$$\Delta x_{\text{CG}} = -50.444 - (-50.789) = +0.35 \text{ in}$$

The longitudinal CG moves 0.35 in forward. Moving the accumulator back pulls the CG rearward, but moving the inverter forward and taking 24.6 lb out of the two parts, which both sit behind the CG, has a bigger effect and pushes it forward.

3.4 Effect on Weight Distribution

With the ARG26 wheelbase of 61 in, a 0.35 in forward shift moves about 0.6% of the weight onto the front axle:

$$\Delta(\text{front \%}) = 100 \times 0.35 / 61 = 0.6\%$$

So a car that was 50/50 would become about 50.6/49.4 front to rear.

4 Remedy: Moving the Axles

For 50/50, the CG has to sit at the midpoint between the front and rear axles. Since the CG moves forward 0.35 in, we move both the front and rear axles forward 0.35 in. The midpoint moves with them, the car goes back to 50/50, and the wheelbase stays at 61 in.

A Appendix: Input Sources

Input	Value	Source
ARG26 sprung mass with driver	470 lb	Sprung mass without the driver, 320 lb, plus a driver weight of 150 lb.
ARG26 sprung CG, x	-50.79 in	The car is designed to 50/50, so the CG is half the wheelbase behind the front axle: $-20.289 - 61 \times 0.5 = -50.789$. Front axle location from CAD (Figure 3).
ACC26 mass	102 lb	ARG26 weight budget, accumulator current weight of 46.266 kg (Figure 10).
ACC26 CG, x	-64.00 in	iProperties of the accumulator master assembly in CAD, -1625.284 mm (Figure 4).
INV26 mass	34.17 lb	ARG26 weight budget, inverter current weight of 15.5 kg (Figure 11).
INV26 CG, x	-78.20 in	iProperties of the inverter enclosure assembly in CAD (Figure 5).
ACC27 mass	95 lb	Estimate from Aryaman (HV lead).
ACC27 CG, x	-69.04 in	Rectangular approximation of the accumulator. The ARG27 case is longer at the front, so its CG moves forward by half the added length, +2.56 in (Figure 6). The whole accumulator also moves back 7.60 in to fill the space behind it that the ARG26 inverter took up (Figure 7, Figure 8). Total: $+2.56 - 7.60 = -5.04$ in from the ACC26 CG.
INV27 mass	16.53 lb	Estimate of 7.5 kg for the new inverter from Aryaman (HV lead).
INV27 CG, x	-74.77 in	Assumes the same enclosure with its CG in the middle, turned from standing up on ARG26 to lying flat on ARG27. The CG moves forward by half the height minus half the width, $13.01 / 2 - 6.15 / 2 = +3.43$ in, from the INV26 CG (Figure 9).
Wheelbase, L	61 in	From our design. Kept the same on ARG27.

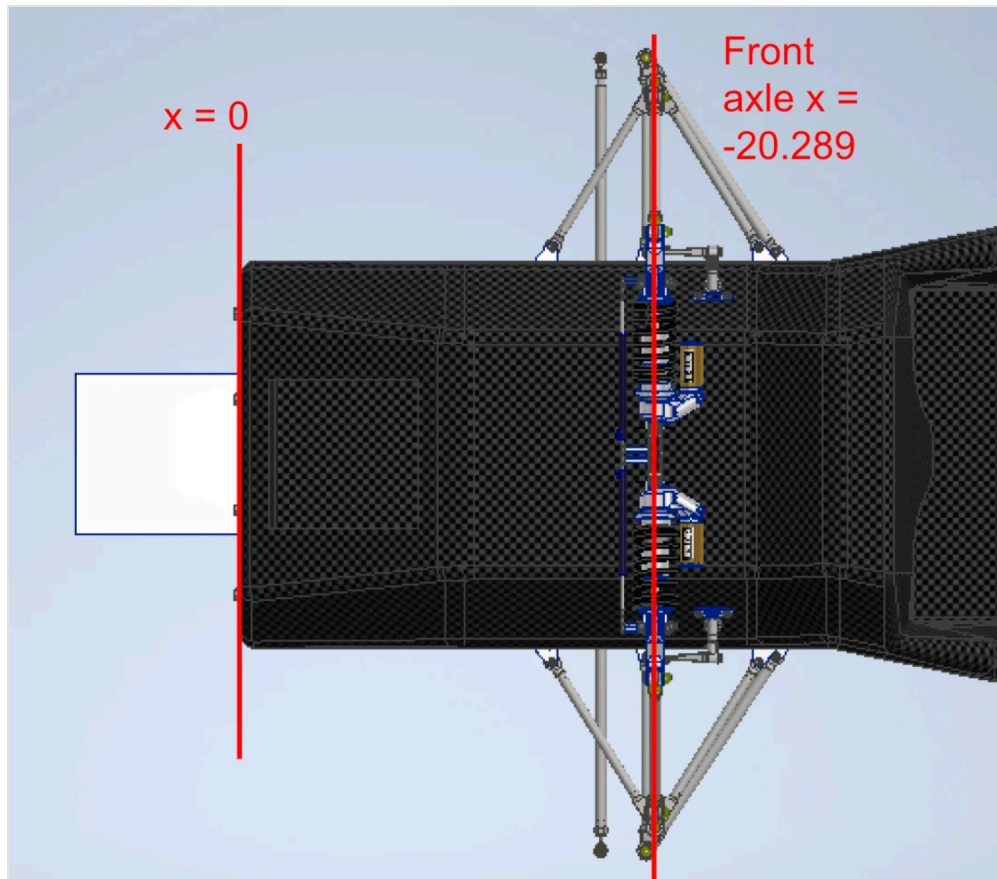


FIGURE 3 ARG26 front axle location in CAD, $x = -20.289$ in.

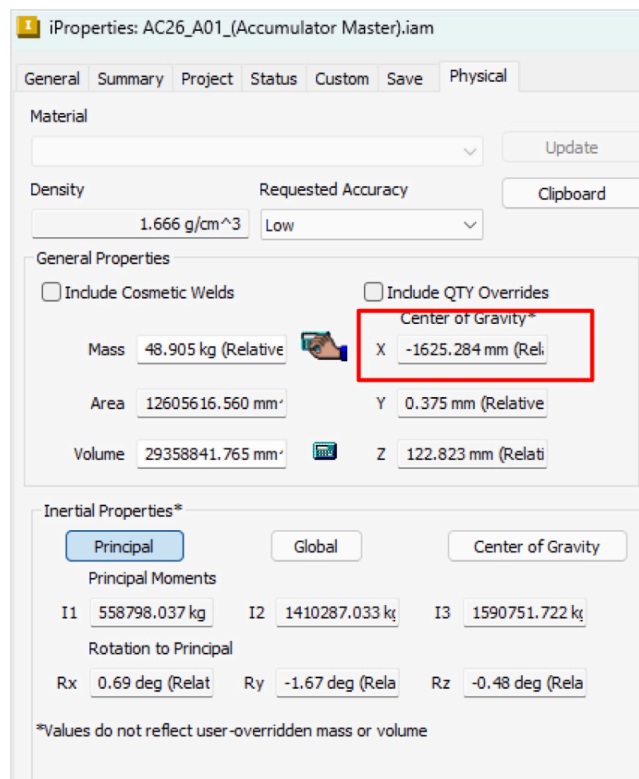


FIGURE 4 iProperties of the ARG26 accumulator master assembly, CG $x = -1625.284$ mm (-64.00 in).

iProperties: PH26_A01_(Inverter enclosure).iam

General Summary Project Status Custom Save Physical

Material

Density 9.569 g/cm³ Requested Accuracy Low

General Properties

☐ Include Cosmetic Welds ☐ Include QTY Overrides

Center of Gravity*

Mass 151.776 lbmass (Rel) X -78.200 in (Relative)

Area 6944.263 in² (Rel) Y 0.140 in (Relative E)

Volume 439.032 in³ (Rel) Z 6.230 in (Relative E)

Inertial Properties*

Principal Global Center of Gravity

Principal Moments

I1 4971.367 lbmas I2 2007.530 lbmas I3 3598.761 lbmas

Rotation to Principal

Rx -1.41 deg (Rela) Ry -1.16 deg (Rela) Rz -0.07 deg (Rela)

*Values do not reflect user-overridden mass or volume

FIGURE 5 iProperties of the ARG26 inverter enclosure assembly, CG x = -78.20 in.

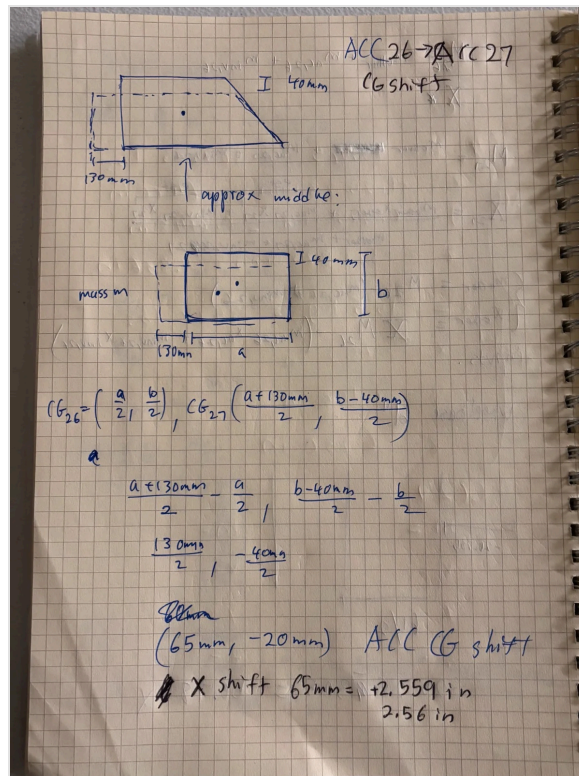


FIGURE 6 Hand calculation of the ACC27 CG shift from the longer case, using a rectangular approximation of the accumulator.

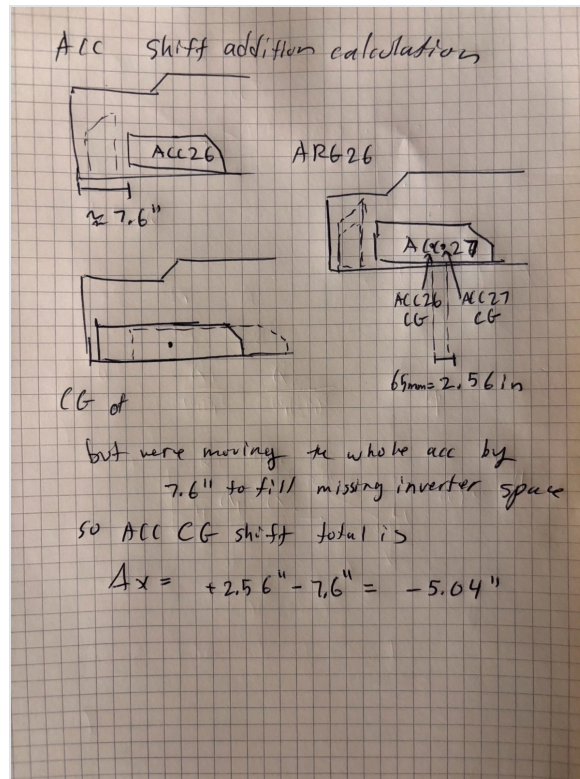


FIGURE 7 Continuation of Figure 6. The whole accumulator moves back 7.6 in to fill the inverter space, so the total ACC CG shift is $+2.56 - 7.6 = -5.04$ in.

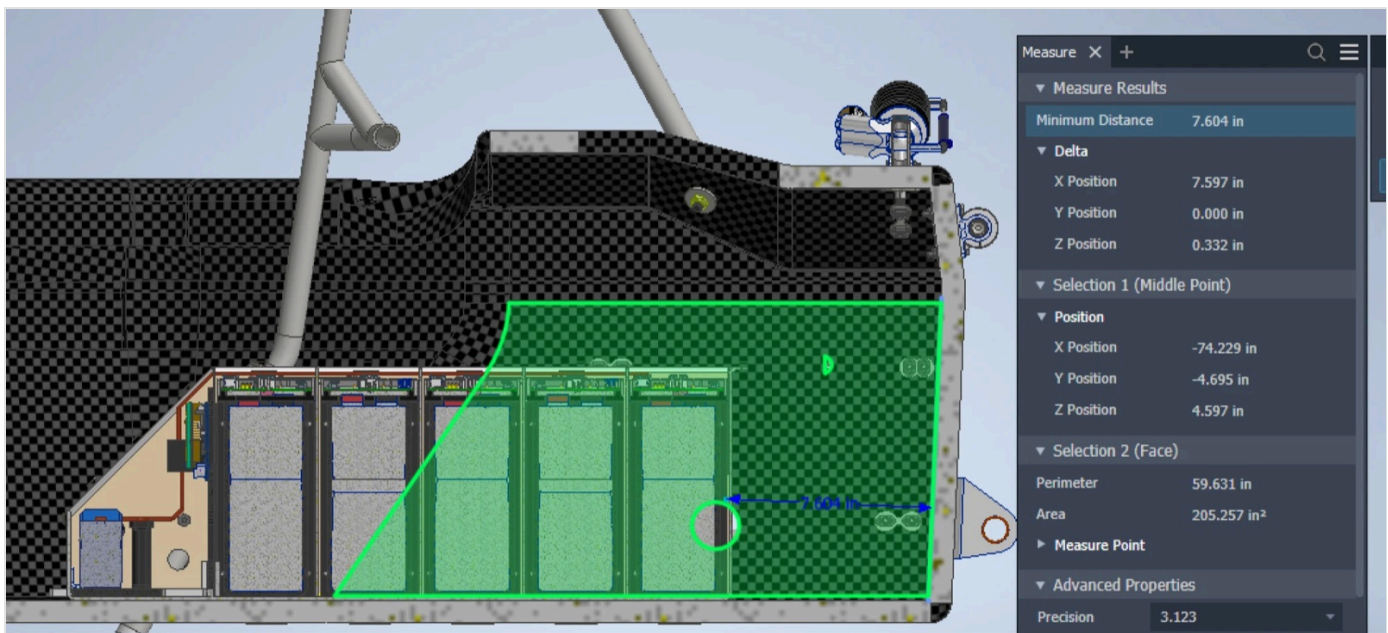


FIGURE 8 Space behind the ARG26 accumulator measured in CAD, 7.597 in in x (rounded to 7.6 in).

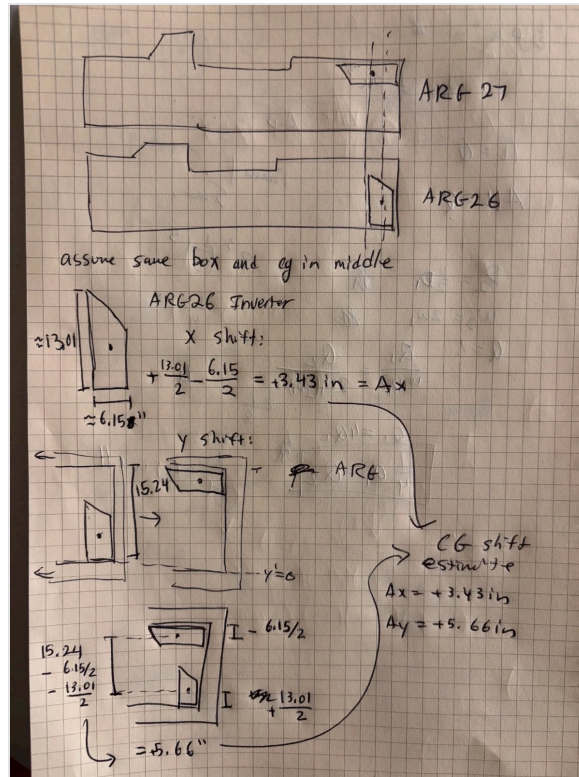


FIGURE 9 Hand calculation of the INV27 CG shift, $x = +3.43 \text{ in}$.

Accumulator Weight Targets							
Component	Current Wt. on ARG26 [kg]	ARG25 Wt. [kg]	Target Weight for ARG26 [kg]	TARGET ARG26 Wt. to ARG26 Wt. Delta [kg]	ARG25 to TARGET ARG26 Wt. Delta [kg]	ARG25 to CURRENT ARG26 Wt. Delta [kg]	Notes
Lower Case	8.680	7.801	4.000	4.680	-3.801	0.879	
Case Lid	0.000	1.860	1.000	-1.000	-0.860	-1.860	
Segments	34.500	37.013	37.000	-2.500	-0.013	-2.513	
Acc Electronics	2.523	2.495	2.200	0.323	-0.295	0.028	
Maintenance Plugs	0.563	1.000	0.800	-0.237	-0.200	-0.437	
TOTAL	46.266	50.169	45.000	1.266	-5.169	-3.903	

FIGURE 10 ARG26 weight budget, accumulator weight targets.

Powertrain Peripherals Weight Targets							
Component	Current Wt. on ARG26 [kg]	ARG25 Wt. [kg]	Target Weight for ARG26 [kg]	TARGET ARG26 Wt. to ARG26 Wt. Delta [kg]	ARG25 to TARGET ARG26 Wt. Delta [kg]	ARG25 to CURRENT ARG26 Wt. Delta [kg]	Notes
Inverters	15.500	15.513	14.000	1.500	-1.513	-0.013	enclosure with inverters, connectors, and current shunt
Radiators	2.500	2.500	2.300	0.200	-0.200	0.000	2x 1.25kg radiators
Radiator Fans	1.640	1.500	1.300	0.340	-0.200	0.140	2x 0.83kg fan
Cooling Lines and Fittings	1.580	4.000	3.500	-1.920	-0.500	-2.420	
Pump	1.996	1.500	1.350	0.646	-0.150	0.496	2x 2.2lb pumps
Catch Can	0.050	0.050	0.050	0.000	0.000	0.000	
Non-acc HVD		0.150	0.150	0.000	0.000	-0.150	
HV Cables	2.506	3.000	2.700	-0.194	-0.300	-0.494	
Water	1.424	2.000	2.000	-0.576	0.000	-0.576	.824L (motor/inv. loop tubing) +2(rad water vol (aprox .3L) + 4(sleeve
LV Box	0.340	0.267	0.250	0.090	-0.017	0.073	including telemetry
Nickel Paint	0.100	0.500	0.500	-0.400	0.000	-0.400	
Harness	3.500	4.540	4.400	-0.900	-0.140	-1.040	
Control Panel	0.601	0.584	0.400	0.201	-0.184	0.017	
Dashboard	0.178	0.500	0.400	-0.222	-0.100	-0.322	
TOTAL	31.915	36.604	33.300	-1.013	-3.304	-4.689	

FIGURE 11 ARG26 weight budget, powertrain peripherals weight targets.

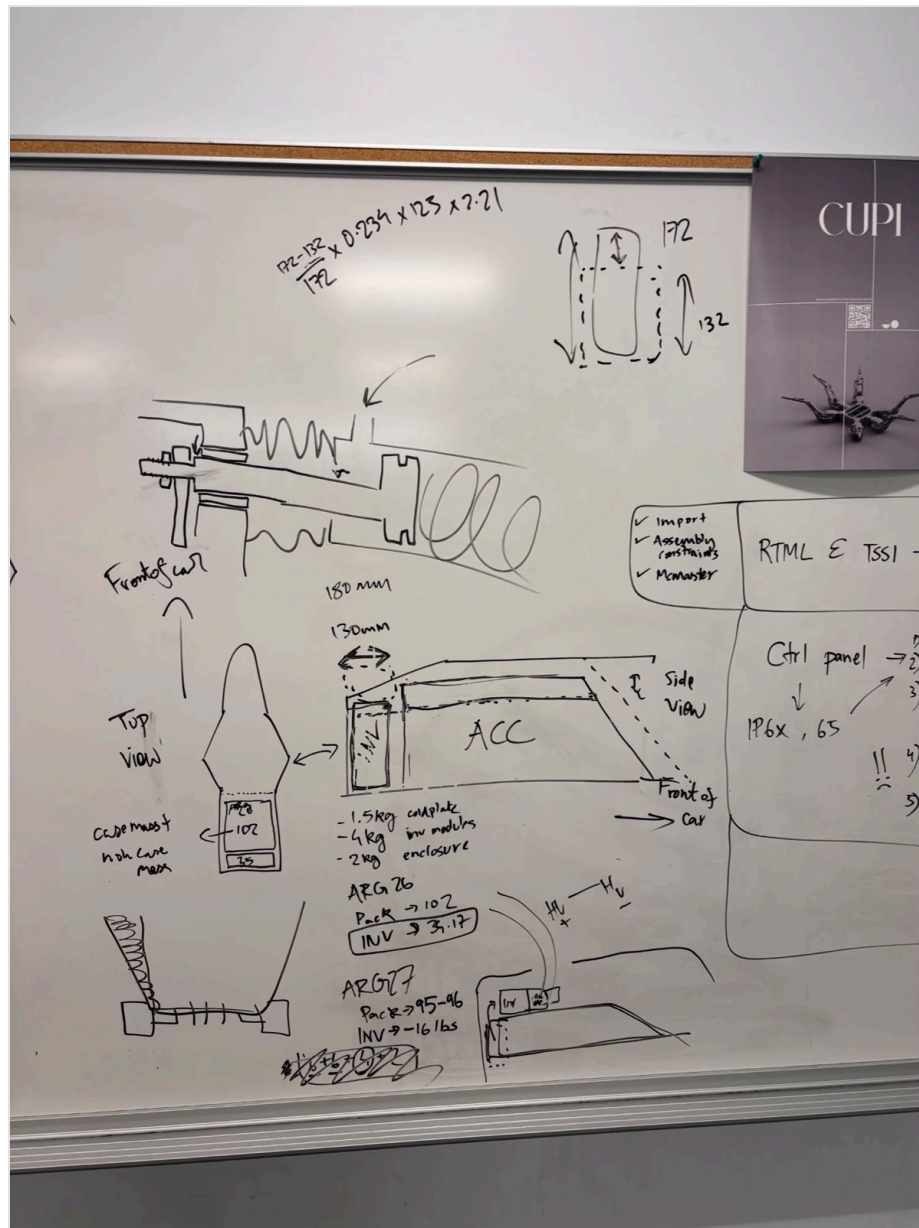


FIGURE 12 Whiteboard sketches mapping out the initial work on the accumulator and inverter changes.

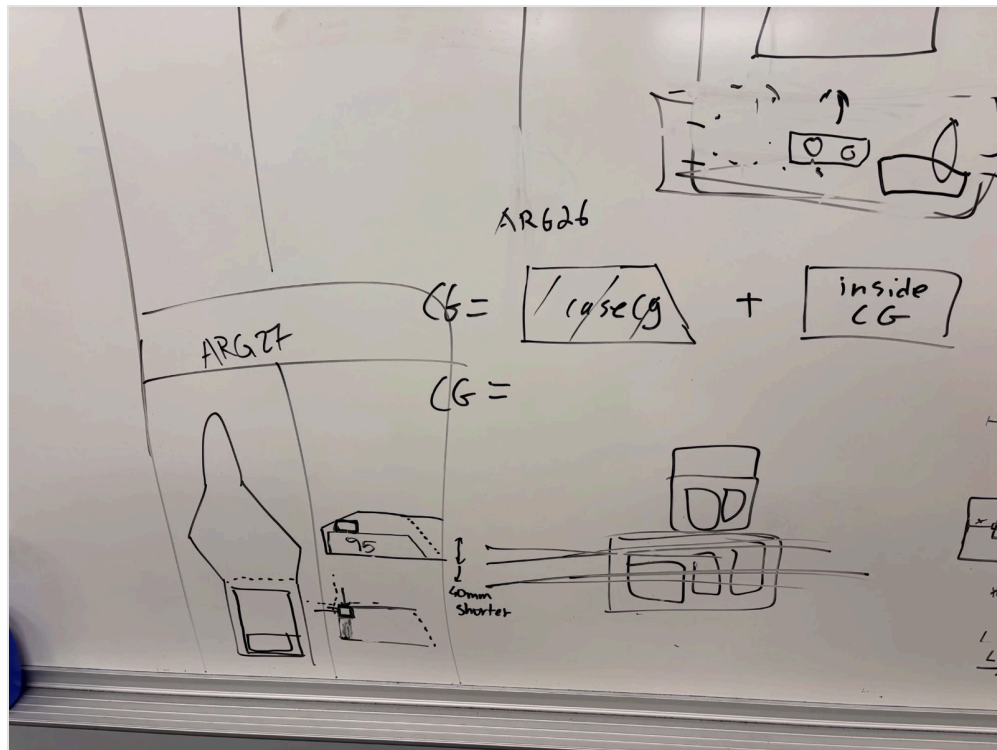


FIGURE 13 Whiteboard sketches comparing the ARG26 and ARG27 accumulator layouts.